



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

SEMESTER 1/20252026

SECJ 3553 - ARTIFICIAL INTELLIGENCE

SECTION 02

PROJECT:

SMART AGRICULTURE SYSTEM
(ASSIGNMENT 3)

THEME:

AGRICULTURE

LECTURER:

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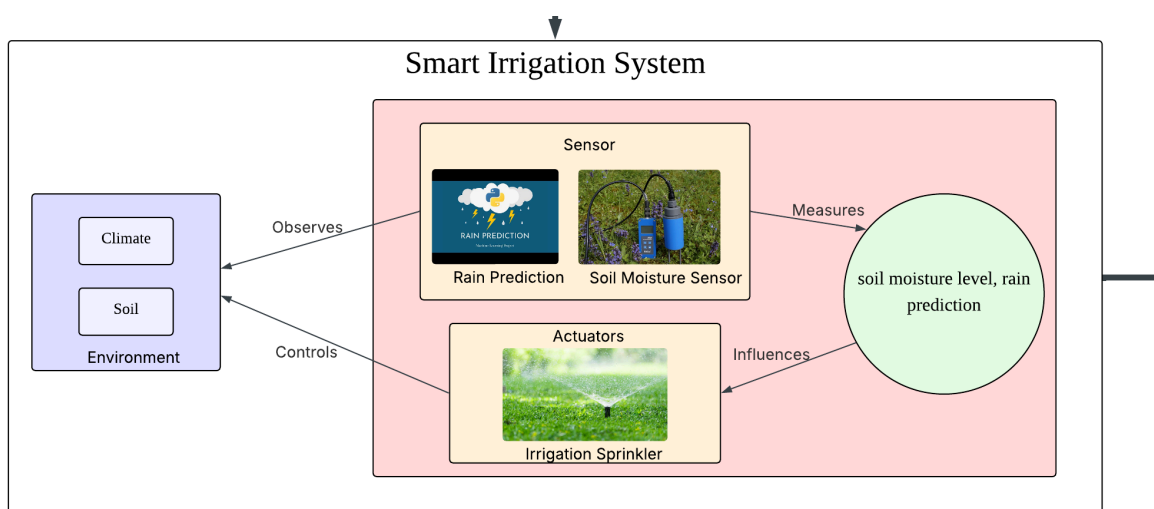
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1.0 PEAS MODEL FOR SUB-AGENTS AND VISUAL DIAGRAMS

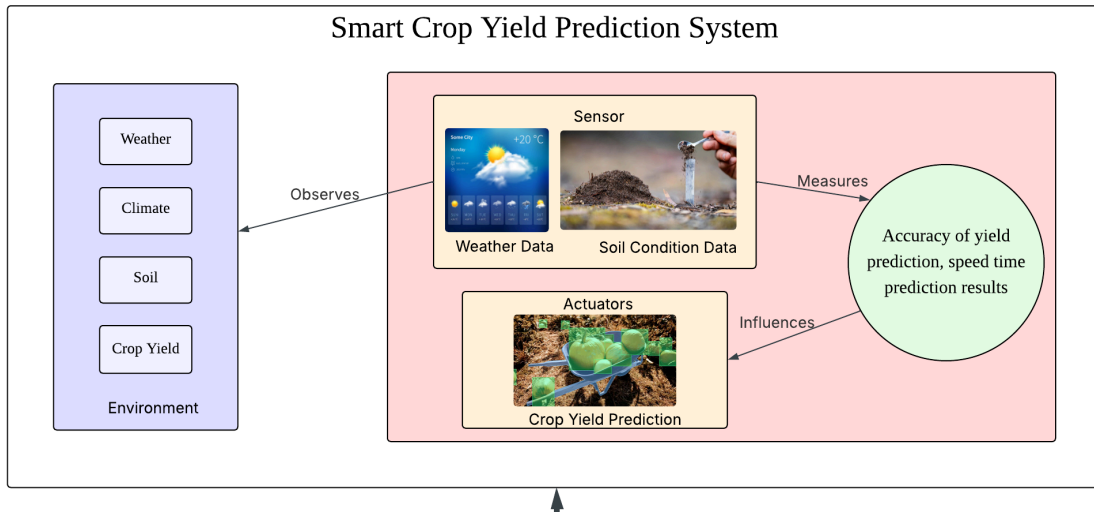
1.1 PEAS Model for Irrigation System

Agent Type	Smart Irrigation System
Performance and Measure	Soil moisture level, rain prediction
Environment	Weather, soil
Actuators	Irrigation sprinkler
Sensors	Soil moisture sensor, weather history data



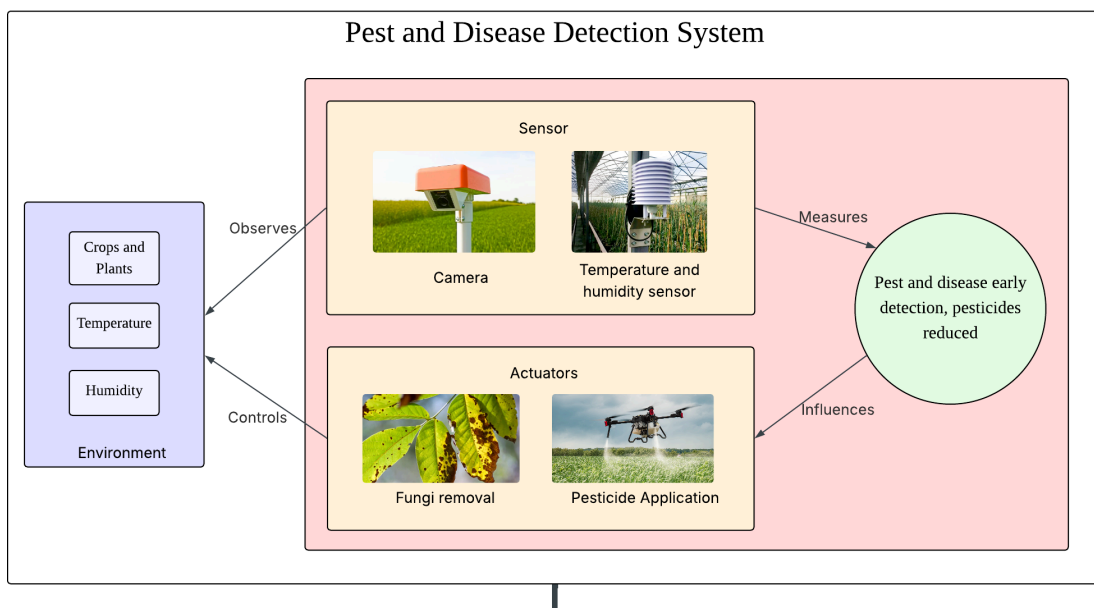
1.2 PEAS Model for Smart Crop Yield Prediction System

Agent Type	Smart Crop Prediction System
Performance and Measure	Accuracy of yield prediction, speed time prediction results
Environment	Weather conditions, climate patterns, soil status, crop growth history
Actuators	Generate prediction of crop yield
Sensors	Weather monitoring data, climate trends, soil condition data, historical performance crop data



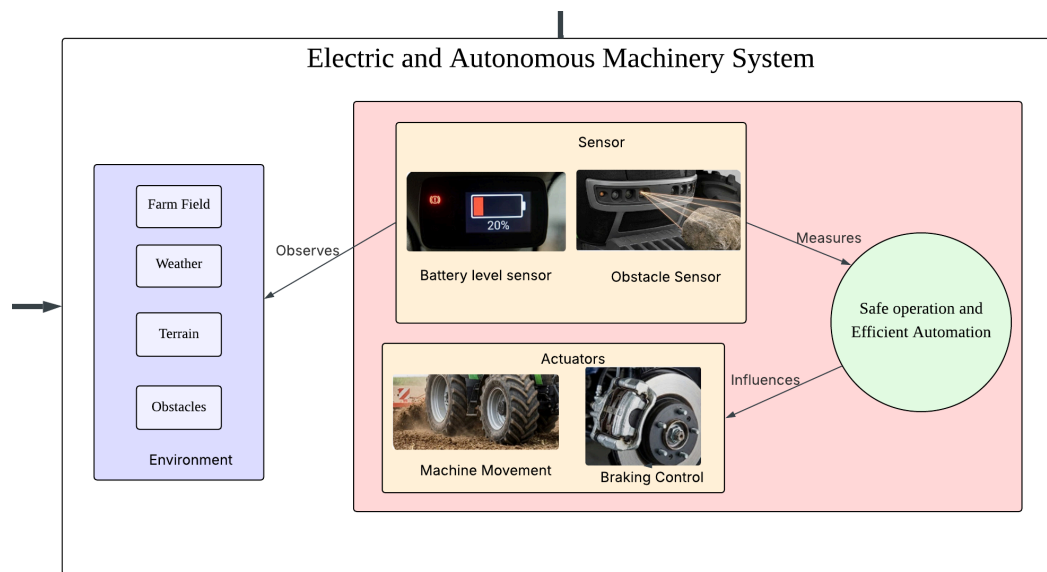
1.3 PEAS Model for Smart Pest and Disease Detection System

Agent Type	Smart Pest and Disease Detection Agent
Performance and Measure	Pest and disease early detection, pesticides reduced
Environment	Crops and plants, temperature, humidity
Actuators	Apply pesticide and fungi removal
Sensors	Camera, temperature sensors, humidity sensors

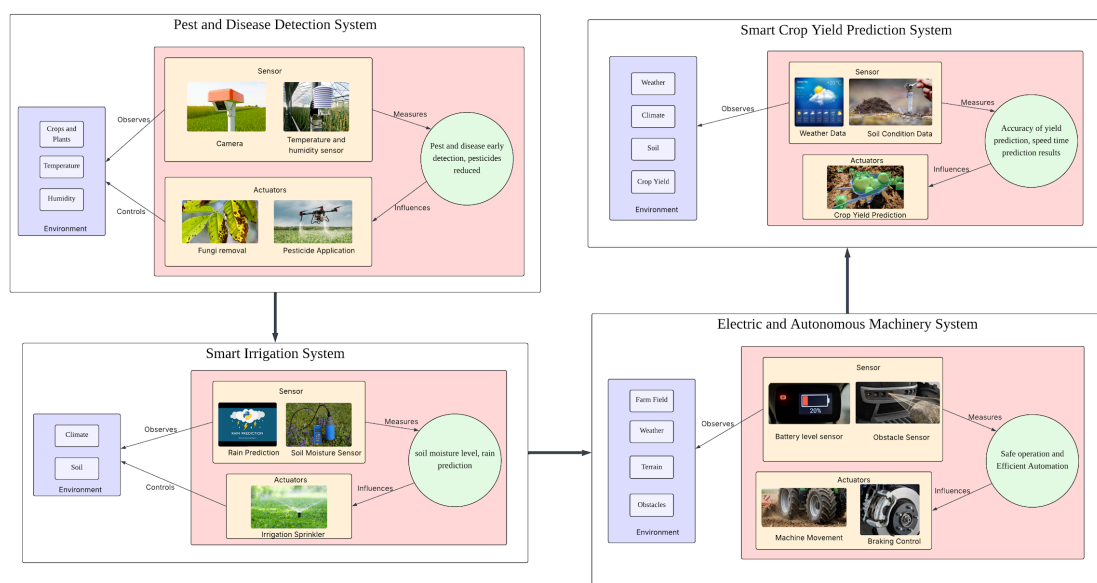


1.4 PEAS Model for Electric and Autonomous Machinery

Agent Type	Electric and Autonomous Machinery
Performance and Measure	Safety, efficiency, collision avoidance, low emissions
Environment	Farm field, weather, soil, terrain, obstacles
Actuators	Machine movement, braking control
Sensors	Battery level sensor, obstacle sensor



1.5 Unified Model Diagram



2.0 PEAS MODEL REPRESENTATION IN PROOF OF CONCEPT (POC)

Performance:

The performance measure serves as the primary goal, which defines success as maximizing agricultural output while reducing the cost of resources like power, water, and pesticides. The precision of autonomous navigation without collisions, the accuracy of disease diagnosis, and the proximity of the expected yield to the actual harvest are all used to quantify success. The prototype guarantees that each action made by the irrigation system or the machinery immediately contributes to the total productivity and sustainability of the farm by aligning these measures.

Environment:

The environment for this system is the physical and digital landscape of the farm. Physically, this includes the kind of soil, the range of crops cultivated, and the unpredictable climate. It also incorporates "dynamic obstacles" that the autonomous machinery must avoid, including farmworkers, wandering animals, or uneven terrain. The yield prediction module must evaluate past climate data and current weather trends from the digital environment.

Actuators:

To interact with and adapt to the surroundings, the system uses a variety of actuators, which are components that do physical or digital work. Mechanical components include the electric motors and steering systems of autonomous tractors, valves that open and close irrigation pipelines, and precision nozzles on drones used for pesticide application and fungal control. On the digital side, the actuators include the user interface or dashboard that operates by displaying the farmer's final crop production forecast. These actuators enable the prototype to transition from passive observation to active intervention, such as shutting off the water when rain is predicted where the weather will provide the required moisture.

Sensors:

The system's sensors, which serve as the data inputs for the whole architecture, are solely responsible for its ability to detect its surroundings. Soil moisture probes and humidity meters give a continuous stream of information on the "soil condition," while high-resolution cameras and optical sensors are utilized for image recognition to identify pests and illnesses early. To guarantee the autonomous machinery functions securely, sensors such as GPS for location tracking and obstacle detectors for obstacle detection manage navigation and safety. Furthermore, the prediction system processes external data that is pulled in via virtual sensors or APIs, such as historical climate records and weather forecasts. When combined, these sensors offer the provided data from the environments that enable the prototype to make informed choices for each of the four modules.